Geometry, Perimeter & Area — Level 2 — Answer Key

1. **C) 124 sq m** — Outer area = 20×15 = 300 sq m. A path 2 m wide running just inside the boundary leaves an inner rectangle of (20−4)×(15−4) = 16×11 = 176 sq m. Path area = 300−176 = 124 sq m.
2. **C) 275 sq m** — A path built outside adds 2.5 m all around, so the outer rectangle becomes (30+5)×(20+5) = 35×25 = 875 sq m. Inner (the park itself) = 30×20 = 600 sq m. Path area = 875−600 = 275 sq m.
3. **C) 600 sq cm** — Outer area = 40×30 = 1,200 sq cm. The photo itself (inner rectangle, 5 cm in from each side) = (40−10)×(30−10) = 30×20 = 600 sq cm. Frame area = 1,200−600 = 600 sq cm.
4. **C) 26 sq m** — Total area = 5×4 + 3×2 = 20+6 = 26 sq m (the two sections' areas simply add, regardless of exactly how they're joined).
5. **C) 87 sq cm** — Original area = 12×8 = 96 sq cm. Removing the 3×3 corner square removes 9 sq cm. Remaining area = 96−9 = 87 sq cm.
6. **B) 2 times** — Perimeter = 2(l+b). Doubling both: new perimeter = 2(2l+2b) = 2×2(l+b) = 2× the original perimeter.
7. **C) 9 times** — Area = side². Tripling the side: new area = (3×side)² = 9×side² = 9× the original area.
8. **C) 48 cm** — Five 4 cm squares in a row form a rectangle 20 cm long (5×4) and 4 cm wide. Perimeter = 2(20+4) = 48 cm.
9. **C) 90 sq cm** — Each tile has area 5×3 = 15 sq cm. Six tiles placed with no gaps or overlaps always cover a total area of 6×15 = 90 sq cm, however they're arranged.
10. **D) 975 sq m** — Field area = 40×25 = 1,000 sq m. Flower bed area = 5×5 = 25 sq m. Uncovered area = 1,000−25 = 975 sq m.
11. **C) 32 cm** — The original rectangle's perimeter is 2(10+6) = 32 cm. Cutting a rectangular notch out of a corner removes two short stretches of the original boundary but adds two new cut edges of exactly the same total length back in — so the perimeter of the L-shape is unchanged: still 32 cm.
12. **C) 4** — The plus-sign shape has a horizontal line of symmetry, a vertical line of symmetry, and both diagonal lines of symmetry — 4 in total.
13. **C) 70°** — The angles of a triangle always add up to 180°. x = 180 − 50 − 60 = 70°.
14. **C) 26** — The full grid has 6×5 = 30 unit squares. Removing a 2×2 block removes 4 unit squares. Remaining area = 30−4 = 26 square units.
15. **C) 42 cm** — Area 49 sq cm means each tile has side 7 cm (since 7×7 = 49). Two tiles side by side form a rectangle 14 cm by 7 cm. Perimeter = 2(14+7) = 42 cm.

How to use this: 10+/15 is strong for a Level-2-focused set. Q11 (the corner-notch perimeter) is the trickiest idea here — it's worth re-deriving with pencil and paper: sketch the rectangle, cut the corner,

and measure each remaining side yourself rather than just trusting the shortcut.